



RealEye

Your eye for what's real.

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CSMI 4071



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The Problem

As AI-generated images, videos, emails, and text become increasingly realistic and widespread online, it is becoming harder for everyday users, especially those unfamiliar with AI, to distinguish between authentic and AI-generated content. This makes people more vulnerable to scams and impersonation. It also spreads misinformation and makes it harder to know what to trust online.

People correctly distinguished AI-generated people from real people only 61% of the time.

The 2024 meta-analysis found that interventions including AI assistance, feedback training, and other detection strategies increased deepfake detection performance to about 65.14%, compared with detection that was otherwise around chance level.

AI misinformation is already a major public concern. In a Pew Research Center survey of 9,720 U.S. adults, **57% said they were extremely or very concerned that AI would be used to create and distribute fake or misleading information**

Beautiful apartment. Amazing price. One small problem... it doesn't exist.

Susan, a 70 year old woman, is looking for a new apartment. She finds an unbelievably perfect rental listing and thinks she has found the deal of a lifetime. The only problem? ***The house was generated by AI and doesn't exist.***

***Luxury beachfront apartment
Only \$800/month
Pay the deposit today!***

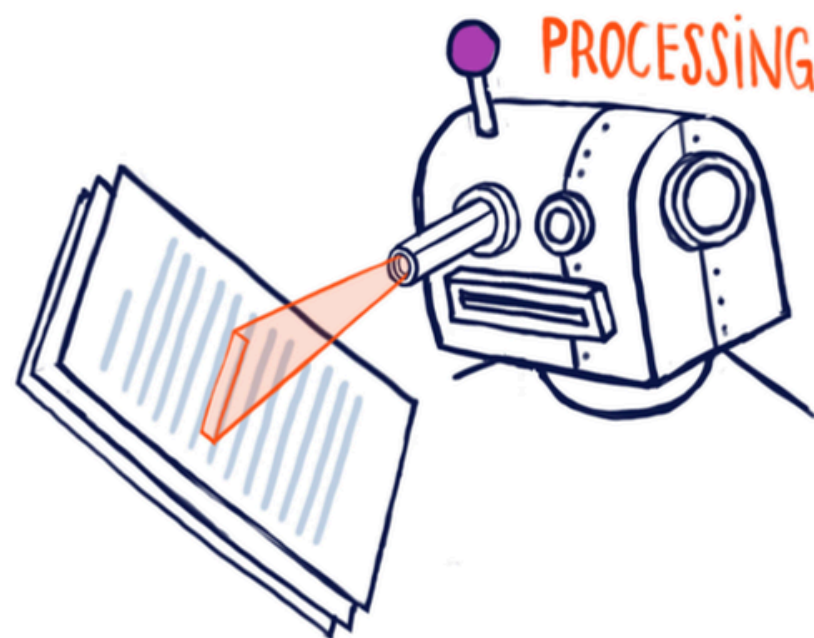


**AI-GENERATED IMAGES AND
CONVINCING MESSAGES CAN TURN
A FAKE LISTING INTO A REAL
FINANCIAL LOSS.**

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AI Content Detector

A browser/mobile tool that helps everyday users identify potentially AI-generated content while browsing, messaging, or checking email.



How it works

1. Encounter Content

The user installs our Chrome extension and browses the web as usual – viewing emails, messages, images, videos, and other online content.

2. Automatic Detection

With the click of a button, the extension scans the content on the current webpage and analyzes it using detection techniques.

3. Clear Visual Warning

Potentially AI-generated content is visually flagged; for example, with a red highlight/border and a “Possibly AI-Generated” warning.

4. More Information

Users can click the warning to see additional information, such as the type of content detected and the system's confidence level.

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Synth ID

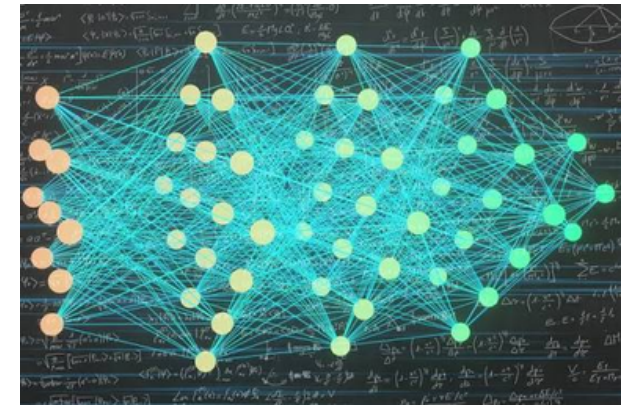
SynthID is an **invisible watermark** developed by Google DeepMind that is embedded in AI-generated text, images, audio, and video. Google uses it across its generative AI products, and other companies such as OpenAI and Anthropic have begun incorporating SynthID into AI-generated media.



C2PA

The Coalition for Content Provenance and Authenticity, or C2PA, provides an open technical standard for publishers, creators, and consumers to establish the **origin and edits of digital content**. The Content Credentials pin reveals the method of creation and allows users to see a record of editing history.



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How to Build the Detector

For **text/email**, we use a machine-learning classifier trained on examples of human-written and AI-generated text. It can look at patterns such as sentence structure, predictability, repetition, vocabulary, and writing style.

For **images**, we can use an image-classification model trained on real vs. AI-generated images. It may learn visual artifacts that commonly appear in generated images, such as unusual textures, inconsistent details, or patterns in pixels that humans do not easily notice.

Content → determine content type → send to appropriate detection model → receive probability

Train the Machine Learning Model

1. Build a **dataset** containing human-generated examples and AI-generated examples
2. **Label everything** as either 0 = Human or 1 = AI-generated
3. Use an **existing pretrained model** as the base, such as a pretrained vision model or transformer model, and fine-tune it rather than building a neural network entirely from scratch.
4. **Train the final classification layer** on your data – during training, it sees labeled examples and adjusts its weights to improve predictions.

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How Our Detection System Works

Our software uses multiple detection methods rather than relying on a single indicator.

1. Check for SynthID

Scan the content for an embedded SynthID watermark.

If detected → strong evidence of AI-generated content.

2. Check C2PA / Content Credentials

Look for verified provenance information about how the content was created or edited.

If AI generation is recorded → evidence of AI-generated content.

3. Run Our Machine Learning Model

Regardless of whether provenance information is available, analyze the content itself using our trained text or image classifier.

Model → estimated probability that the content is AI-generated.

4. Show the Results

Combine the available evidence to determine an overall result and display it directly on the webpage.

By combining verified provenance signals with machine-learning analysis, our system can detect known AI-generated content while still evaluating content that does not contain a watermark or Content Credentials

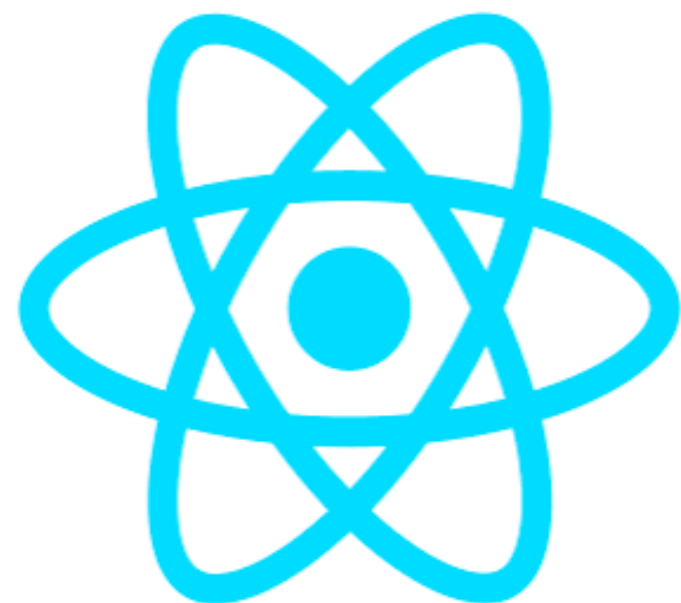


Home

Photo

About Us

Framework



REACT



**REST API +
CHROME
EXTENSION APIS**



**Hugging Face
Transformers**

Hugging Face Transformers
provides APIs and tools to
download, train, and fine-tune pre-
trained machine learning models



SYNTH ID

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What Makes RealEye Different?

- **Layered and transparent analysis:** Separately displays the machine-learning prediction, SynthID result, and available C2PA Content Credentials, allowing users to understand the evidence behind the result.
- **Designed for older adults:** Provides readable warnings, simple explanations, and guided verification steps for users with limited technical experience.
- **Trusted Circle verification:** Lets users share suspicious content and its analysis with a trusted family member for a second opinion.





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Why RealEye Is a Good Fit for This Course

Builds on what we know

Web development • APIs • Machine Learning • UI/UX

Pushes us further

Chrome extensions • AI detection • model integration • content provenance

Challenging, but realistic

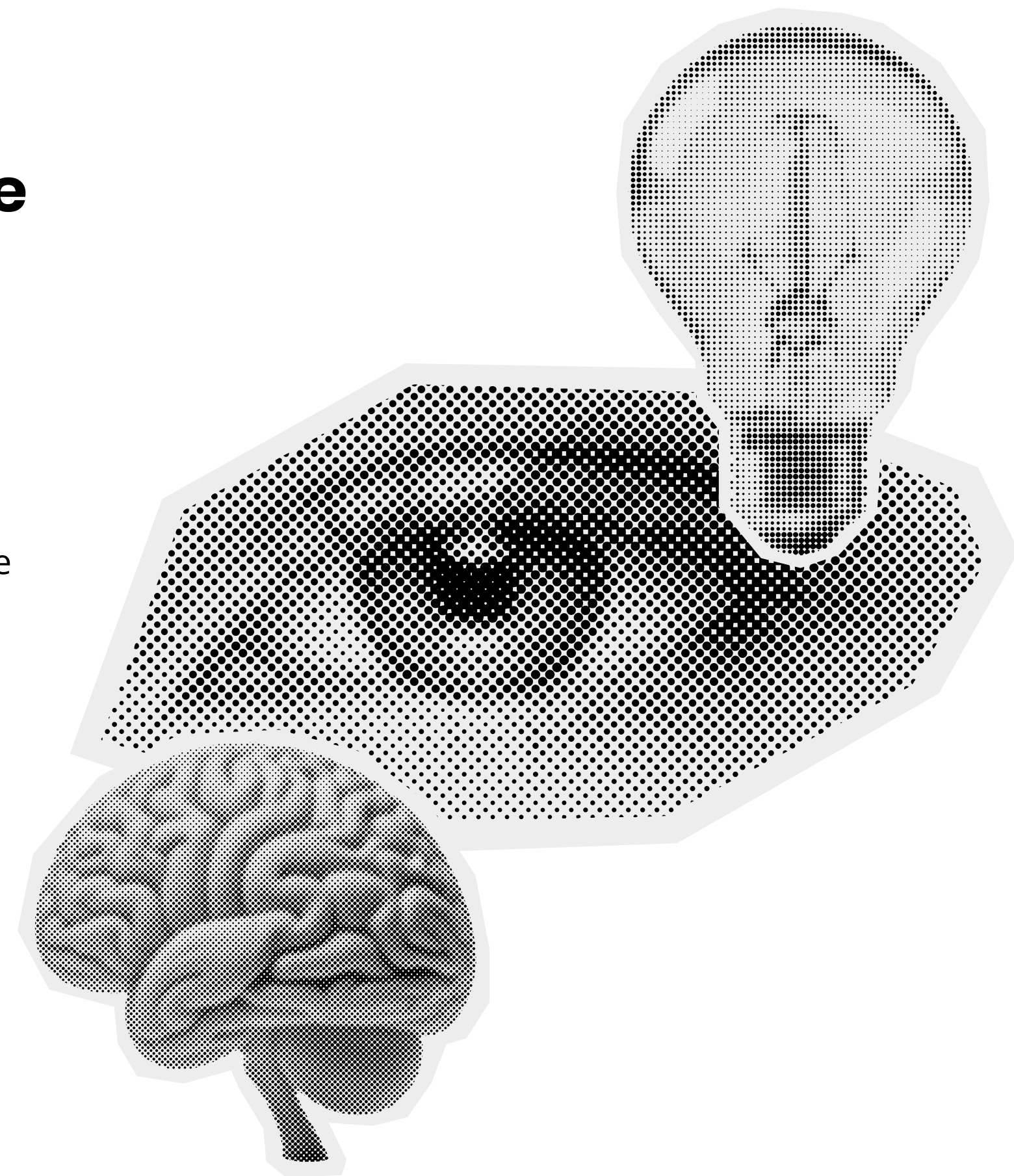
Focused core features + optional stretch goals

Designed for a semester

Start with text/image detection → expand only if time allows

Meaningful application

Addresses a growing problem with AI-generated content



Core Goals/Features

Functional Chrome Extension that can scan the current webpage

Check for AI watermark/C2PA provenance signals where technically accessible

Detect AI-generated text + images using a pretrained/fine-tuned ML model

Generate an AI likelihood/confidence score

Highlight/flag detected content directly on the webpage

Evaluate detector accuracy using a held-out test dataset

Stretch Goals

Video/deepfake detection

Scam/phishing detection

Fact-checking / misinformation detection

User feedback on incorrect classifications



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Project Scope/Goals





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Thank You